Problem 1

Implement a **Frame-based knowledge system** to represent information about **Employees in an organization**.

Requirements

1. Create a base frame Employee with slots:
    - name
    - empId
    - department
    - salary
2. Create a derived frame Manager that:
    - Inherits all slots from Employee
    - Adds an extra slot teamSize
3. Allow the user to:
    - Enter employee details
    - Display inherited and specific slot values

Sample Input

Enter role: Manager
Enter Name: Ramesh
Enter Employee ID: E102
Enter Department: IT
Enter Salary: 85000
Enter Team Size: 10

Sample Output

Frame: Manager
Name: Ramesh
Employee ID: E102
Department: IT
Salary: 85000
Team Size: 10
(Inherited from Employee frame)

Problem 2

Implement a **Semantic Network** to represent knowledge about **Animals** and perform **inheritance-based reasoning**.

Requirements

1. Represent the following concepts:
   - Animal, Bird, Mammal, Sparrow, Dog
2. Define relationships:
   - is-a
   - can
3. Store facts:
   - Bird can fly
   - Mammal can walk
4. Answer user queries using inheritance:
   - "Can a Sparrow fly?"
   - "Can a Dog walk?"

Sample Input

Enter Query:
Can Sparrow fly?

Sample Output

Yes
Reason: Sparrow → Bird → can fly

Another Sample Input

Enter Query:
Can Dog walk?

Sample Output

Yes
Reason: Dog → Mammal → can walk